

## PADILLA BAY RESERVE, WA, USA

WMO: 998007

Lat: 48.467N

Lon: 122.467W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|------------|-------------|------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |            | 1%          |            |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB       | WS          | MCDB       | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-5.2 | (c)<br>-3.2 | (d)<br>-10.0                  | (e)<br>1.6 | (f)<br>-3.0 | (g)<br>-7.3 | (h)<br>2.0 | (i)<br>-1.4 | (j)<br>12.0           | (k)<br>8.3 | (l)<br>10.9 | (m)<br>8.3 | (n)<br>2.0            | (o)<br>60 | (p)<br>0.563 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD<br>to 0.4% DB |          |
|---------------|---------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------------|----------|
|               |                           | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                         |          |
|               |                           | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                    | PCWD     |
| (a)<br>8      | (b)<br>10.1               | (c)<br>23.8     | (d)<br>18.2 | (e)<br>22.5 | (f)<br>17.5 | (g)<br>21.4 | (h)<br>16.9 | (i)<br>19.0         | (j)<br>22.8 | (k)<br>18.1 | (l)<br>21.6 | (m)<br>17.4 | (n)<br>20.7 | (o)<br>3.7              | (p)<br>0 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.3                            | 12.4 | 21.1 | 16.6 | 11.8 | 20.1 | 15.7 | 11.2 | 19.3 | 53.6          | 22.8 | 50.9 | 21.8 | 48.7 | 20.8 | 22.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |  |  |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|--|--|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |  |  |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |  |  |
| (a) 10.0          | (b) 8.4 | (c) 7.4 | (e) -8.0                   | (f) 26.9 | (g) 2.2            | (h) 1.5 | (i) -9.6                                           | (j) 28.0 | (k) -10.9  | (l) 28.9 | (m) -12.2  | (n) 29.7 | (o) -13.8  | (p) 30.8 |  |  |
|                   |         |         | DB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |
|                   |         |         | WB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |
|                   |         |         |                            |          |                    |         |                                                    |          |            |          |            |          |            |          |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.7    | 4.2   | 4.6   | 6.7   | 8.6   | 11.6  | 14.2  | 15.9  | 16.0  | 14.0  | 10.1  | 6.4   | 4.5  |
|                                                                                  | DBStd       | 5.10   | 3.25  | 3.51  | 2.73  | 2.13  | 2.11  | 2.04  | 1.80  | 1.75  | 2.25  | 2.83  | 3.55  | 3.41 |
|                                                                                  | HDD10.0     | 826    | 181   | 153   | 108   | 53    | 8     | 0     | 0     | 0     | 1     | 35    | 114   | 174  |
|                                                                                  | HDD18.3     | 3142   | 438   | 386   | 362   | 293   | 209   | 126   | 78    | 76    | 130   | 256   | 358   | 430  |
|                                                                                  | CDD10.0     | 734    | 1     | 1     | 4     | 9     | 57    | 125   | 183   | 184   | 122   | 38    | 7     | 2    |
|                                                                                  | CDD18.3     | 7      | 0     | 0     | 0     | 0     | 0     | 1     | 3     | 2     | 1     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 56     | 0     | 0     | 0     | 0     | 3     | 8     | 28    | 15    | 2     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 2      | 0     | 0     | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.9    | 3.2   | 3.1   | 3.6   | 3.2   | 2.8   | 2.6   | 2.3   | 2.3   | 2.5   | 2.9   | 3.2   | 3.2  |
| Precipitation                                                                    | PrecAvg     | 1105   | 135   | 87    | 104   | 90    | 79    | 64    | 29    | 36    | 55    | 112   | 170   | 142  |
|                                                                                  | PrecMax     | 1427   | 228   | 157   | 161   | 132   | 145   | 118   | 80    | 117   | 121   | 214   | 328   | 212  |
|                                                                                  | PrecMin     | 856    | 73    | 13    | 37    | 25    | 27    | 13    | 2     | 3     | 17    | 33    | 64    | 73   |
|                                                                                  | PrecStd     | 150    | 38    | 36    | 30    | 23    | 30    | 30    | 18    | 30    | 32    | 51    | 65    | 34   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.3  | 13.8  | 15.7  | 17.7  | 21.3  | 24.4  | 25.7  | 25.0  | 23.0  | 19.3  | 15.3  | 14.2 |
|                                                                                  |             | MCWB   | 10.3  | 11.1  | 11.7  | 13.2  | 15.5  | 18.4  | 19.9  | 18.8  | 16.9  | 15.7  | 13.2  | 12.4 |
|                                                                                  | 2%          | DB     | 11.0  | 11.6  | 13.7  | 15.5  | 19.2  | 22.5  | 24.0  | 23.4  | 21.2  | 17.2  | 13.7  | 11.8 |
|                                                                                  |             | MCWB   | 9.3   | 9.5   | 10.3  | 11.5  | 14.9  | 17.2  | 18.5  | 18.0  | 16.8  | 14.5  | 11.8  | 10.3 |
|                                                                                  | 5%          | DB     | 10.0  | 10.4  | 12.3  | 14.0  | 17.8  | 20.6  | 22.5  | 22.1  | 19.8  | 15.7  | 12.6  | 10.3 |
|                                                                                  |             | MCWB   | 8.4   | 8.6   | 9.8   | 10.4  | 14.1  | 16.1  | 17.6  | 17.5  | 16.1  | 13.6  | 11.0  | 8.7  |
|                                                                                  | 10%         | DB     | 8.8   | 9.3   | 11.1  | 12.7  | 16.4  | 19.2  | 21.3  | 21.0  | 18.6  | 14.7  | 11.4  | 9.2  |
|                                                                                  |             | MCWB   | 7.6   | 7.8   | 9.0   | 9.6   | 13.1  | 15.0  | 17.0  | 16.9  | 15.6  | 12.9  | 10.1  | 7.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 10.7  | 11.9  | 12.5  | 13.8  | 16.6  | 19.6  | 20.9  | 19.8  | 18.8  | 16.6  | 13.7  | 12.7 |
|                                                                                  |             | MCDB   | 11.6  | 13.6  | 14.6  | 17.1  | 19.6  | 23.9  | 24.9  | 23.8  | 21.3  | 18.7  | 14.7  | 14.1 |
|                                                                                  | 2%          | WB     | 9.8   | 10.0  | 10.8  | 12.2  | 15.4  | 17.9  | 19.0  | 18.8  | 17.5  | 15.3  | 12.2  | 10.6 |
|                                                                                  |             | MCDB   | 10.8  | 11.2  | 12.9  | 14.6  | 18.8  | 21.6  | 23.1  | 22.5  | 20.0  | 16.7  | 13.4  | 11.8 |
|                                                                                  | 5%          | WB     | 8.8   | 9.0   | 10.1  | 11.0  | 14.4  | 16.5  | 18.0  | 18.0  | 16.7  | 14.1  | 11.2  | 9.1  |
|                                                                                  |             | MCDB   | 9.8   | 10.1  | 12.0  | 13.1  | 17.4  | 20.0  | 22.0  | 21.4  | 19.0  | 15.3  | 12.3  | 10.1 |
|                                                                                  | 10%         | WB     | 7.8   | 8.1   | 9.1   | 10.1  | 13.3  | 15.4  | 17.2  | 17.3  | 16.0  | 13.1  | 10.2  | 8.1  |
|                                                                                  |             | MCDB   | 8.8   | 9.2   | 11.0  | 12.2  | 15.9  | 18.7  | 21.0  | 20.6  | 18.1  | 14.5  | 11.3  | 9.2  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.0   | 5.8   | 6.8   | 7.6   | 8.4   | 9.0   | 10.4  | 10.1  | 8.5   | 7.2   | 6.1   | 4.7  |
|                                                                                  |             | MCDBR  | 5.1   | 6.5   | 7.9   | 9.8   | 11.0  | 11.6  | 12.4  | 11.7  | 10.5  | 7.8   | 6.0   | 4.7  |
|                                                                                  |             | MCWBR  | 4.0   | 4.8   | 5.4   | 6.3   | 7.1   | 6.6   | 7.5   | 6.8   | 6.2   | 5.1   | 4.5   | 3.6  |
|                                                                                  | 5% WB       | MCDBR  | 5.0   | 5.4   | 7.5   | 8.8   | 10.5  | 11.0  | 12.1  | 11.2  | 9.4   | 5.9   | 5.3   | 4.6  |
|                                                                                  |             | MCWBR  | 4.1   | 4.2   | 5.2   | 6.0   | 7.0   | 6.7   | 7.6   | 6.9   | 5.7   | 4.5   | 4.3   | 3.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.315  | 0.312 | 0.327 | 0.350 | 0.373 | 0.355 | 0.347 | 0.352 | 0.338 | 0.330 | 0.319 | 0.312 |      |
|                                                                                  | taud        | 2.535  | 2.561 | 2.536 | 2.457 | 2.411 | 2.475 | 2.524 | 2.518 | 2.552 | 2.566 | 2.542 | 2.499 |      |
|                                                                                  | Ebn at Noon | 764    | 858   | 896   | 901   | 889   | 906   | 908   | 890   | 872   | 822   | 751   | 713   |      |
|                                                                                  | Edh at Noon | 60     | 72    | 86    | 103   | 112   | 106   | 99    | 96    | 84    | 71    | 59    | 55    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.93   | 1.81  | 2.83  | 4.29  | 5.48  | 6.00  | 6.56  | 5.53  | 3.94  | 2.17  | 1.11  | 0.75  |      |
|                                                                                  | RadStd      | 0.12   | 0.24  | 0.33  | 0.40  | 0.43  | 0.57  | 0.46  | 0.36  | 0.32  | 0.24  | 0.17  | 0.10  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (69 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page